

Simple Games

- Games: a mathematical formalism for rational interaction

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- Games: a mathematical formalism for rational interaction
- What is the most popular solution concept? (Nash Equilibrium)

Types of Uncertainty

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Alleatory

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Markov Decision Process

Types of Uncertainty

Alleatory

Epistemic (Static)



Markov Decision Process

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Markov Decision Process

Reinforcement Learning

Types of Uncertainty

Alleatory



Markov Decision Process

Epistemic (Static)



Reinforcement Learning

Epistemic (Dynamic)



POMDP

Types of Uncertainty

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Interaction

Types of Uncertainty

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POMDP

Interaction



Game

Normal Form Games

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		S	W
A	S	4	2
	W	3	1

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Called a **Normal Form, Simple**, or **Bimatrix** Game

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Question for today: What **solution concept** should we use for games?

Dominant Strategies

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Definitions

- Action $a^i \in A^i$
- Joint Action $a = (a^1, \dots, a^k)$
- All Other Actions
 $a^{-i} = (a^1, \dots, a^{i-1}, a^{i+1}, \dots, a^k)$
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Deterministic Best Response:

Action a^i is a deterministic best response
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$$R^i(a^i, a^{-i}) \geq R^i(a^{i'}, a^{-i}) \quad \forall a^{i'}$$

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- **Dominant (Pure) Strategy:** Action a is a dominant strategy if it is a best response to every action taken by the other player.
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Is the dominant strategy equilibrium always the best outcome for the players?

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- Each can either keep silent (**S**) or testify (**T**)
 - other keeps silent → minor conviction (1 year)
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Player 1

Player 2

	S	T
S	-1, -1 → -4, 0	-4, 0
T	0, -4 → -3, -3	-3, -3

DSE

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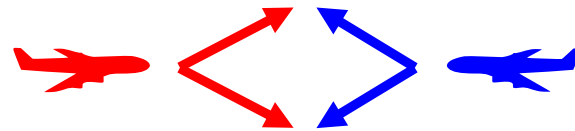
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Do all simple games have a dominant strategy equilibrium?

Collision Avoidance Game

Collision Avoidance Game

Example: Airborne Collision Avoidance



Player 1

Player 2

Up

Down

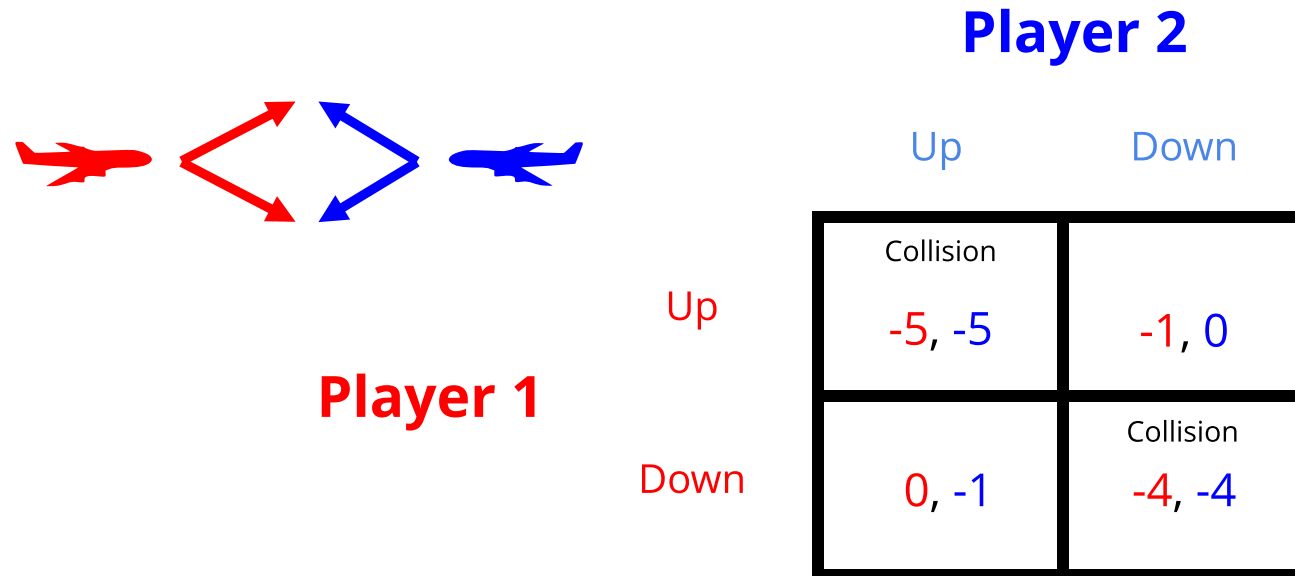
Collision -5, -5	-1, 0
0, -1	Collision -4, -4

Up

Down

Collision Avoidance Game

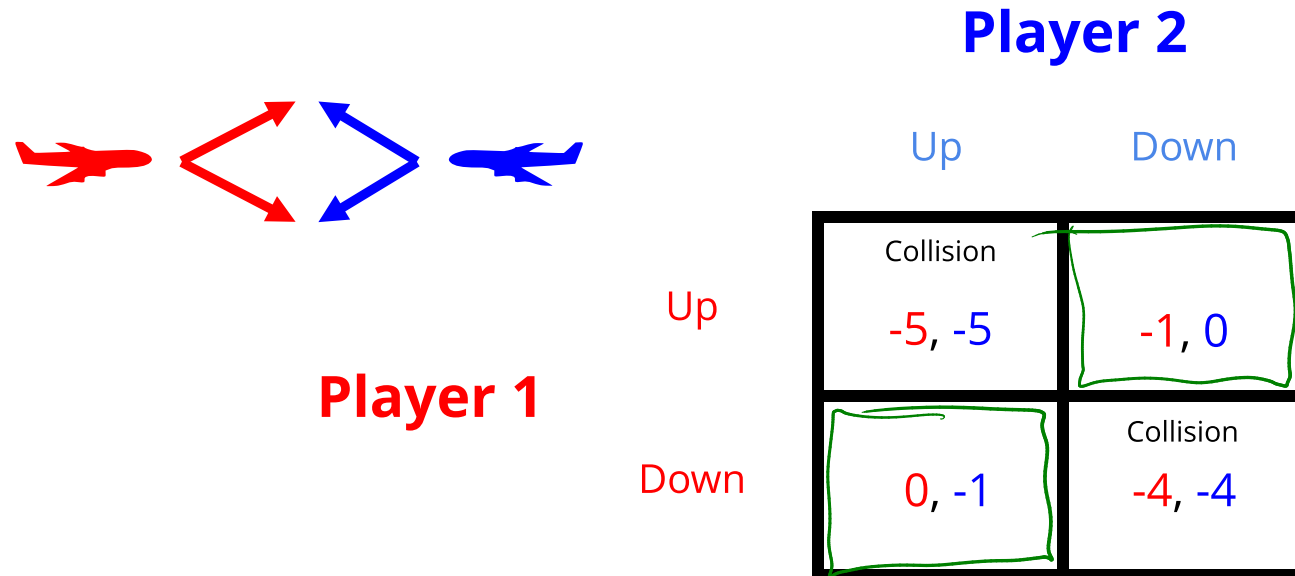
Example: Airborne Collision Avoidance



Pure Nash Equilibrium: All players play a deterministic best response.

Collision Avoidance Game

Example: Airborne Collision Avoidance



Pure Nash Equilibrium: All players play a deterministic best response.

Which equilibrium is better?

Do all simple games have a pure Nash equilibrium?

Practice: Find Pure Nash Equilibria

	a	b	c
a	2	1	10
b	1	2	10
c	10	10	0

Player 1

Player 2

	a	b	c
a	4,4	2,5	0,0
b	5,2	3,3	0,0
c	0,0	0,0	10,10

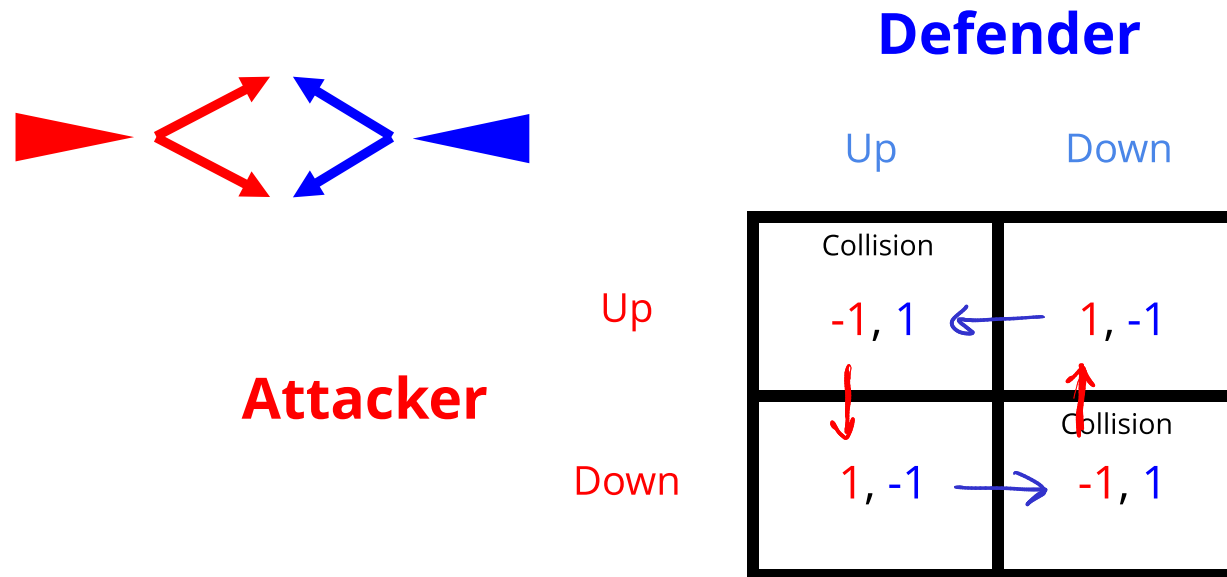
10,3

4,4

Missile Defense Game

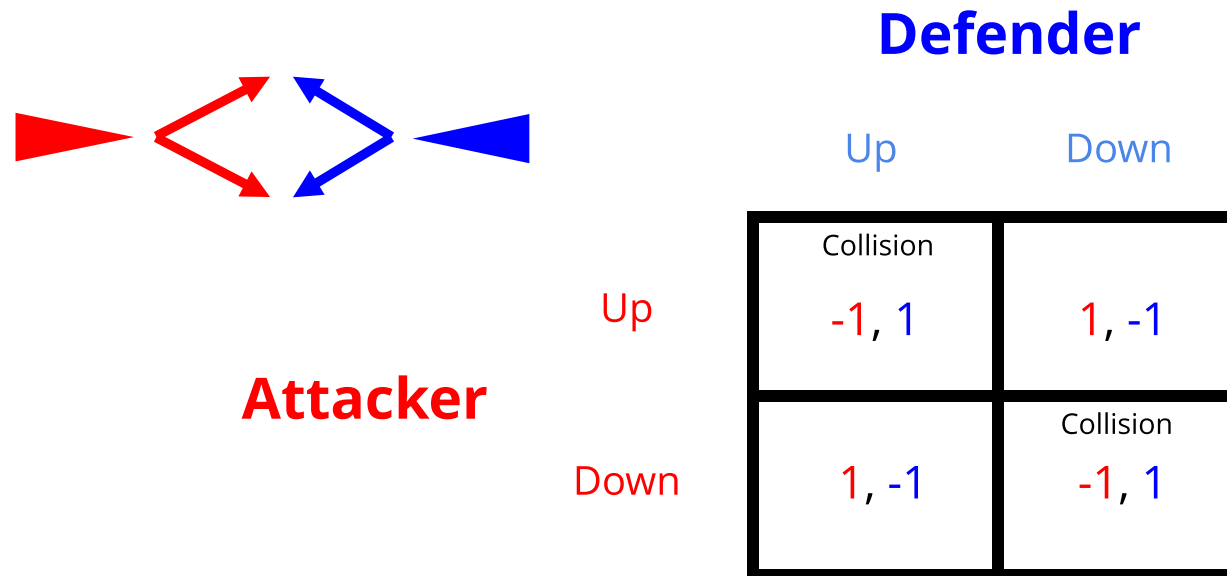
Missile Defense Game

Missile Defense (simplified)



Missile Defense Game

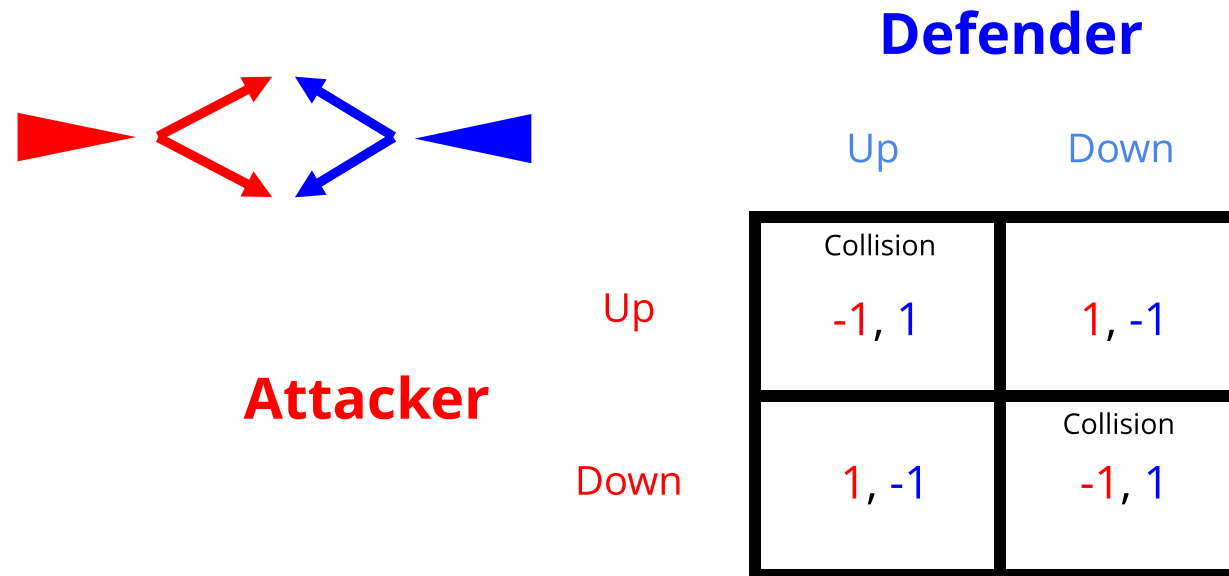
Missile Defense (simplified)



No Pure Nash Equilibrium!

Missile Defense Game

Missile Defense (simplified)



No Pure Nash Equilibrium!

Need a broader solution concept: Mixed Nash equilibrium.

Vocabulary and Notation for Mixed Strategies

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Single Player

Joint

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Two Player Zero Sum:

$$R^1(a) + R^2(a) = 0 \quad \forall a$$

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Cooperative:

$$R^i(a) = R^j(a) \quad \forall i, j, a$$

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("General sum" used to indicate that the game is not limited to zero-sum or coop.)

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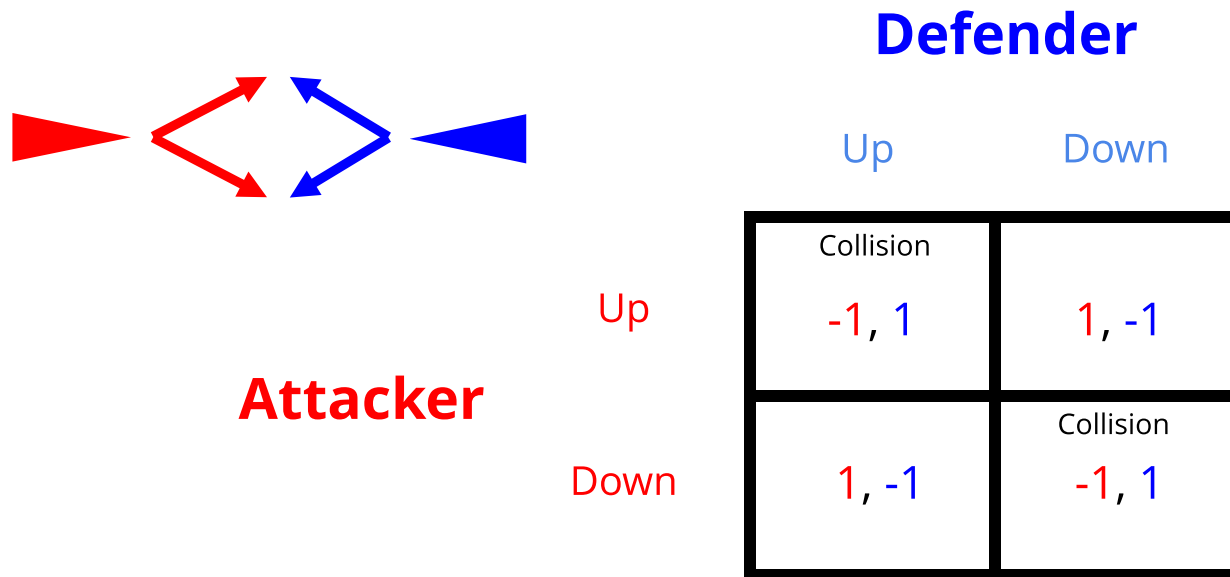
Best Response: Given a joint policy of all other agents, π^{-i} , a best response is a policy π^i that satisfies

$$U^i(\pi^i, \pi^{-i}) \geq U^i(\pi^{i'}, \pi^{-i})$$

for all other $\pi^{i'}$.

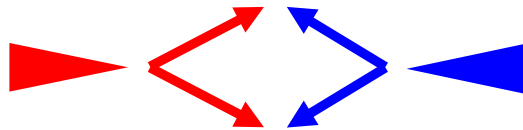
Missile Defense Game

Missile Defense (simplified)



Missile Defense Game

Missile Defense (simplified)



Attacker

Up

Down

Defender

Up

Down

Collision -1, 1	1, -1
1, -1	Collision -1, 1

- A ***Nash equilibrium*** is a joint policy in which **all agents play a best response**

Rock-paper scissors

1. Guess the Nash Equilibrium argument
2. Make a qualitative argument that this is an NE based on best responses

		agent 2		
		rock	paper	scissors
agent 1	rock	0, 0	-1, 1	1, -1
	paper	1, -1	0, 0	-1, 1
	scissors	-1, 1	1, -1	0, 0

NE: each player plays each action with $\frac{1}{3}$ prob.

$$U^2(\pi) = U^2(\pi^1, \pi^2) = \sum_{a^1, a^2} \pi^1(a^1) \pi^2(a^2) R^2(a^1, a^2)$$

$$= \pi^1(r) \pi^2(r)(0) + \pi^1(r) \pi^2(p)(1) + \dots$$

9 combinations

$$U^2(\pi^1, \pi^2) = \pi^2(p) \underbrace{(\pi^1(r) - \pi^1(s))}_{>0} + \pi^2(r) \underbrace{(-\pi^1(p) + \pi^1(s))}_{<0} + \pi^2(s) \underbrace{(\pi^1(p) - \pi^1(r))}_{<0}$$

If $\pi^1(r) > \frac{1}{3}$, $\pi^1(p) = \pi^1(s)$,

PZ's BR is $\pi^2(p) = 1$

Any policy is a BR to a $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ strategy
 A deterministic policy is the BR to any strategy other than $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

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Do all simple games have at least one Nash equilibrium?

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Do all simple games have at least one Nash equilibrium?

Yes!! (might be mixed)

Every finite game has a Nash Equilibrium

Not on Exam

Every finite game has a Nash Equilibrium

EQUILIBRIUM POINTS IN N -PERSON GAMES

By JOHN F. NASH, JR.*

PRINCETON UNIVERSITY

Communicated by S. Lefschetz, November 16, 1949

One may define a concept of an n -person game in which each player has a finite set of pure strategies and in which a definite set of payments to the n players corresponds to each n -tuple of pure strategies, one strategy being taken for each player. For mixed strategies, which are probability distributions over the pure strategies, the pay-off functions are the expectations of the players, thus becoming polylinear forms in the probabilities with which the various players play their various pure strategies.

Any n -tuple of strategies, one for each player, may be regarded as a point in the product space obtained by multiplying the n strategy spaces of the players. One such n -tuple counters another if the strategy of each player in the countering n -tuple yields the highest obtainable expectation for its player against the $n - 1$ strategies of the other players in the countered n -tuple. A self-countering n -tuple is called an equilibrium point.

The correspondence of each n -tuple with its set of countering n -tuples gives a one-to-many mapping of the product space into itself. From the definition of countering we see that the set of countering points of a point is convex. By using the continuity of the pay-off functions we see that the graph of the mapping is closed. The closedness is equivalent to saying: if $P_1, P_2, \dots$ and $Q_1, Q_2, \dots, Q_n, \dots$ are sequences of points in the product space where $Q_n \rightarrow Q$, $P_n \rightarrow P$ and Q_n counters P_n then Q counters P .

Since the graph is closed and since the image of each point under the mapping is convex, we infer from Kakutani's theorem¹ that the mapping has a fixed point (i.e., point contained in its image). Hence there is an equilibrium point.

In the two-person zero-sum case the "main theorem"² and the existence of an equilibrium point are equivalent. In this case any two equilibrium points lead to the same expectations for the players, but this need not occur in general.

* The author is indebted to Dr. David Gale for suggesting the use of Kakutani's theorem to simplify the proof and to the A. E. C. for financial support.

¹ Kakutani, S., *Duke Math. J.*, 8, 457-459 (1941).

² Von Neumann, J., and Morgenstern, O., *The Theory of Games and Economic Behaviour*, Chap. 3, Princeton University Press, Princeton, 1947.

Not on Exam

Every finite game has a Nash Equilibrium

Kakutani's fixed-point theorem

A correspondence $f: X \rightarrow X$ has a fixed point (i.e., $\mathbf{x} \in f(\mathbf{x})$ for some $\mathbf{x} \in X$) if all of the following conditions hold.

- (1) X is a non-empty, closed, bounded, and convex set.
- (2) $f(\mathbf{x})$ is non-empty for any $\mathbf{x}$.
- (3) $f(\mathbf{x})$ is convex for any $\mathbf{x}$.
- (4) The set $\{ (\mathbf{x}, \mathbf{y}) \mid \mathbf{y} \in f(\mathbf{x}) \}$ is closed.

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- Let x be a joint strategy, π .
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- A fixed point of BR is a Nash Equilibrium
- The BR operator and policy space for finite games meet the conditions above

Not on Exam

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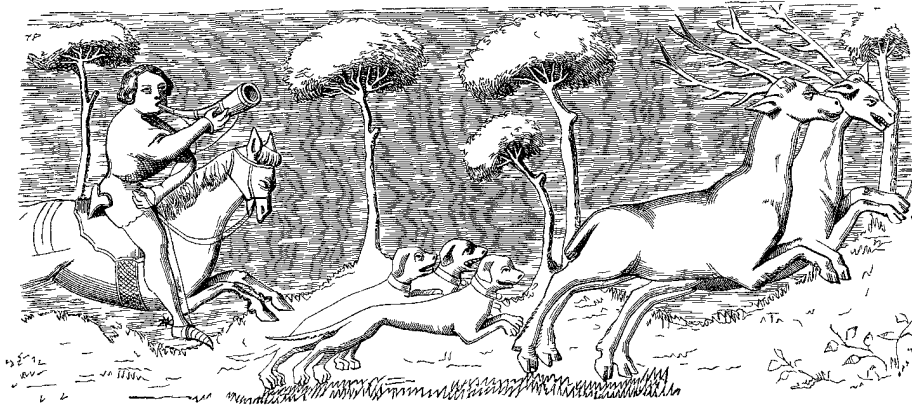
A correspondence $f: X \rightarrow X$ has a fixed point (i.e., $\mathbf{x} \in f(\mathbf{x})$ for some $\mathbf{x} \in X$) if all of the following conditions hold.

- (1) X is a non-empty, closed, bounded, and convex set.
- (2) $f(\mathbf{x})$ is non-empty for any $\mathbf{x}$.
- (3) $f(\mathbf{x})$ is convex for any $\mathbf{x}$.
- (4) The set $\{ (\mathbf{x}, \mathbf{y}) \mid \mathbf{y} \in f(\mathbf{x}) \}$ is closed.

- Let x be a joint strategy, π .
- Let f be BR , that is, the best response operator
- A fixed point of BR is a Nash Equilibrium
- The BR operator and policy space for finite games meet the conditions above
- BR has a fixed point for every finite game, i.e. every finite game has a Nash Equilibrium

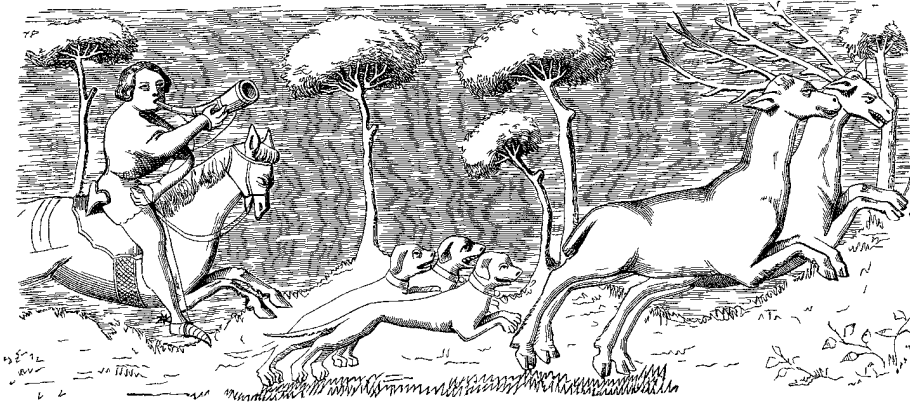
Not on Exam

Calculating Mixed Nash



- In a Mixed Nash Equilibrium, players must be *indifferent* between two or more actions
- (In large games, finding the support of the mixed strategies is the hard part)

Calculating Mixed Nash



Julia Python

	Stag	Hare
Stag	4, 4	1, 3
Hare	3, 1	2, 2

- In a Mixed Nash Equilibrium, players must be *indifferent* between two or more actions
- (In large games, finding the support of the mixed strategies is the hard part)

P2 indifferent

$$\begin{aligned}
 P2: \text{stag} \quad & 4\pi^1(s) + 1\pi^1(H) = v \\
 & 3\pi^1(s) + 2\pi^1(H) = v \\
 & \pi^1(s) + \pi^1(H) = 1 \\
 & \pi^1(s) = \pi^1(H) = 0.5 \quad v = 2.5
 \end{aligned}$$

P1 indifferent

$$P1: \text{stag} \quad 4\pi^2(s) + 1\pi^2(H) = u$$

$$P1: \text{hare} \quad 3\pi^2(s) + 2\pi^2(H) = u$$

$$\pi^2(s) + \pi^2(H) = 1$$

$$\begin{bmatrix} 4 & 1 & -1 \\ 3 & 2 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \pi^2(s) \\ \pi^2(H) \\ u \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned}
 \pi^2(s) &= 0.5 \\
 \pi^2(H) &= 0.5 \\
 u &= 2.5
 \end{aligned}$$

✓ P1's utility

General approach to find Nash Equilibria

General approach to find Nash Equilibria

objective = 0 $\Leftrightarrow$ NE

$$\begin{array}{ll}\text{minimize}_{\pi, U} & \sum_i (U^i - U^i(\pi)) \\ \text{subject to} & \boxed{U^i \geq U^i(a^i, \pi^{-i}) \text{ for all } i, a^i} \quad \text{BR} \\ & \boxed{\sum_{a^i} \pi^i(a^i) = 1 \text{ for all } i} \\ & \boxed{\pi^i(a^i) \geq 0 \text{ for all } i, a^i}\end{array}$$

General approach to find Nash Equilibria

$$\begin{array}{ll} \underset{\pi, U}{\text{minimize}} & \sum_i (U^i - U^i(\pi)) \\ \text{subject to} & U^i \geq U^i(a^i, \pi^{-i}) \text{ for all } i, a^i \\ & \sum_{a^i} \pi^i(a^i) = 1 \text{ for all } i \\ & \pi^i(a^i) \geq 0 \text{ for all } i, a^i \end{array}$$

Not on Exam

Topology of bimatrix games:

	1	6	5	4	3	2	1	6	5	4	3	2
2	2 3 4 1 4 2 Heg. Stability	2 3 4 2 4 3 Samaritan _{su}	2 3 4 2 1 4 Samaritan _{su}	2 3 4 2 1 4 Clock _{su}	2 3 4 2 3 4 Clock _{su}	2 3 4 2 1 4 Endless	2 3 4 2 4 3 Called Bluff	2 3 4 2 4 3 Bully	2 3 4 2 1 4 Unfair	2 3 4 2 4 3 Skewed BoS	2 3 4 2 4 3 Asym BoS	2 3 4 2 4 3 Chicken
3	3 3 4 1 4 2 Samson	3 2 4 1 4 3 Asym Sd _{su}	3 2 4 1 4 3 Asym Sd _{su}	3 2 4 1 4 3 Cycle _{su}	3 2 4 1 4 3 Cycle _{su}	3 3 4 1 4 1 Inspector	3 4 2 1 4 1 Self-serving _{su}	3 4 2 1 4 1 Protector _{su}	3 4 2 1 3 2 Protector _{su}	3 4 2 1 4 3 Favorites _{su}	3 4 2 1 4 3 Battle of Sexes Asym BoS	
4	3 3 4 2 1 4 Dellah	3 2 4 2 1 4 Asym Sd _{su}	3 2 4 2 1 4 Asym Sd _{su}	3 1 4 2 1 4 Pursuit	3 1 4 2 1 4 Pareto	3 2 4 2 1 4 Missile Crisis	3 4 3 2 4 1 Self-serving _{su}	3 4 2 2 4 1 Protector _{su}	3 4 1 2 3 2 Protector _{su}	3 4 1 2 4 3 Hero	3 4 1 2 4 3 Favorites _{su}	3 4 1 2 4 3 Skewed BoS
5	2 3 4 3 4 1 Hostage	2 3 4 3 4 1 Benevolence _{su}	2 3 4 3 4 1 Benevolence _{su}	2 1 4 3 4 1 2nd Best _{su}	2 1 4 3 4 1 2nd Best _{su}	2 1 4 3 4 1 Big Bully	2 3 4 3 4 1 Tragedy	2 3 4 3 4 1 Delight _{su}	2 4 1 3 4 2 Pure Delight	2 3 4 3 4 1 Protector _{su}	2 3 4 3 4 1 Protector _{su}	2 3 4 3 4 1 Unfair
6	1 3 4 3 4 1 Blackmailer	1 2 4 3 4 1 Benevolence _{su}	1 2 4 3 4 1 Benevolence _{su}	1 2 4 3 4 1 2nd Best _{su}	1 2 4 3 4 1 2nd Best _{su}	1 3 4 3 4 1 Hamlet	1 4 3 3 4 1 Total Conflict	1 4 2 3 4 1 Mixed Delight	1 4 2 3 4 1 Delight _{su}	1 4 2 3 4 1 Protector _{su}	1 4 2 3 4 1 Protector _{su}	1 4 2 3 4 1 Bully
1	1 3 4 2 4 1 Id. Hegemony	1 2 4 2 4 1 Samaritan _{su}	1 2 4 2 4 1 Samaritan _{su}	1 1 4 2 4 1 Revelation	1 2 4 2 4 1 Alibi	1 3 4 2 4 1 Asym Pd	1 4 3 2 4 1 Prisoners D.	1 4 2 2 4 1 Total Conflict	1 4 1 2 4 2 Tragedy	1 4 1 2 4 3 Self-serving _{su}	1 4 2 2 4 3 Self-serving _{su}	1 4 3 2 4 2 Called Bluff
2	1 3 4 2 4 1 Win-win	1 2 4 2 4 1 C. Aligned _{su}	1 2 4 2 4 1 C. Aligned _{su}	1 1 4 2 4 1 C. Assurance _{su}	1 2 4 2 4 1 C. Assurance _{su}	1 2 3 2 4 1 Stag Hunt	1 4 4 2 3 1 Asym Pd	1 4 2 2 3 1 Hamlet	1 4 1 2 3 2 Big Bully	1 4 1 2 3 2 Missile Crisis	1 4 2 2 3 1 Inspector	1 4 3 2 3 2 Endless
3	1 3 4 2 1 2 R Assurance _{su}	1 2 4 2 1 2 Commons _{su}	1 2 4 2 1 2 Commons _{su}	1 1 4 2 1 2 Coordination	1 2 4 2 1 2 Coordination	1 3 4 2 1 2 R Assurance _{su}	1 4 4 3 2 1 Alibi	1 4 2 3 2 1 2nd Best _{su}	1 4 1 3 2 2 2nd Best _{su}	1 4 1 3 2 3 Pareto	1 4 2 3 2 3 Cycle _{su}	1 4 3 3 2 2 Clock _{su}
4	2 3 4 3 1 2 R Assurance _{su}	2 3 4 3 1 2 Commons _{su}	2 3 4 3 1 2 Commons _{su}	2 1 4 3 1 2 Coordination	2 1 4 3 1 2 Coordination	2 3 4 3 1 2 R Assurance _{su}	2 4 3 3 2 1 Revelation	2 4 2 3 1 1 2nd Best _{su}	2 4 1 3 1 2 2nd Best _{su}	2 4 1 3 1 3 Pursuit	2 4 2 3 1 3 Cycle _{su}	2 4 3 3 1 2 Clock _{su}
5	3 3 4 2 1 2 Row Aligned _{su}	3 2 4 2 1 3 Harmony _{su}	3 2 4 2 1 3 Harmony-mix	3 1 4 2 1 3 Commons _{su}	3 1 4 2 1 3 Commons _{su}	3 3 4 2 1 3 Row Aligned _{su}	3 4 3 2 1 2 Samaritan _{su}	3 4 2 2 1 2 Benevolent _{su}	3 4 1 2 1 2 Benevolent _{su}	3 4 1 2 1 3 Asym Sd _{su}	3 4 2 2 1 3 Asym Sd _{su}	3 4 3 2 1 2 Samaritan _{su}
6	3 3 4 1 2 1 Row Aligned	3 2 4 1 2 3 Harmony-pure	3 2 4 1 2 3 Harmony	3 1 4 1 2 3 Commons _{su}	3 1 4 1 2 3 Commons _{su}	3 3 4 1 2 1 Row Aligned	3 4 3 1 2 1 Samaritan _{su}	3 4 2 1 2 1 Benevolent _{su}	3 4 1 1 2 2 Benevolent _{su}	3 4 1 1 2 3 Asym Sd _{su}	3 4 2 1 2 3 Asym Sd _{su}	3 4 3 1 2 2 Active Sam _{su}
1	2 3 4 1 3 1 No Conflict	2 3 4 1 3 3 C. Aligned _{su}	2 3 4 1 3 3 C. Aligned _{su}	2 1 4 1 3 3 C. Assurance _{su}	2 1 4 1 3 3 C. Assurance _{su}	2 3 4 1 3 1 Win-win	2 4 3 1 3 1 Id. Hegemony	2 4 2 1 3 1 Blackmailer	2 4 1 1 3 2 Hostage	2 4 1 1 3 3 Dellah	2 4 2 1 3 3 Samson	2 4 3 1 3 2 Heg. Stability
3	3 3 4 1 4 2 Heg. Stability	3 2 4 1 4 2 Samaritan _{su}	3 2 4 1 4 2 Samaritan _{su}	3 2 4 1 4 2 Clock _{su}	3 2 4 1 4 2 Clock _{su}	3 3 4 1 4 2 Endless	3 4 3 1 4 2 Called Bluff	3 4 2 1 4 2 Bully	3 4 1 1 4 2 Unfair	3 4 1 1 4 3 Skewed BoS	3 4 2 1 4 3 Asym	

Algorithms that use best response

Iterated Best Response: randomly cycle between agents who play the best response for the current policy (converges to Nash for certain narrow classes of games)

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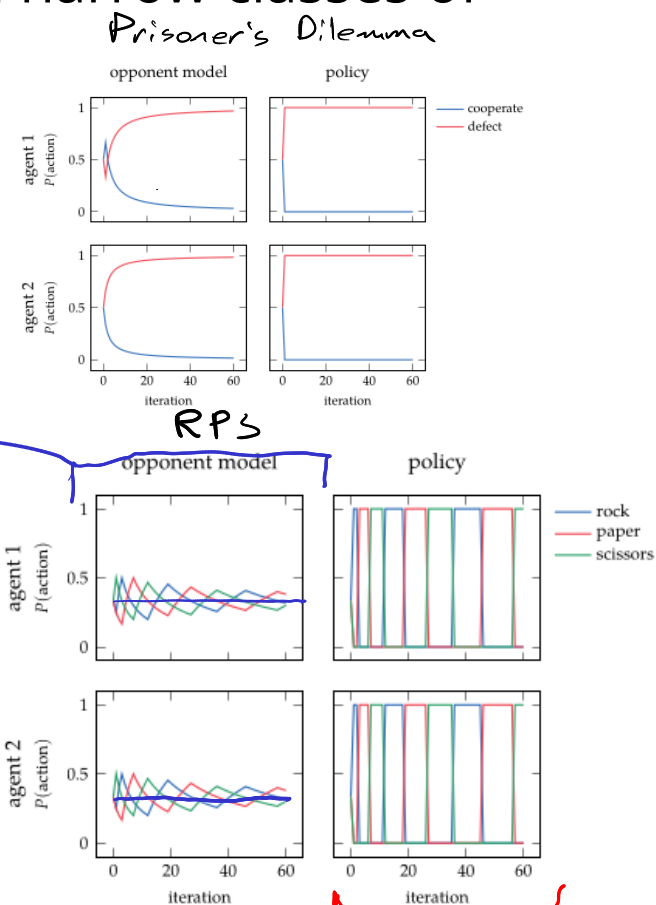
Fictitious Play:

1. Estimate maximum likelihood policies for opponents:

$$\pi^j(a^j) \propto N(j, a^j)$$

2. Play best response to estimated policy

(The joint policy $\pi^j(a^j) \propto N(j, a^j) \quad \forall j, a^j$ converges to Nash games, notably zero-sum)



Bach or Stravinsky

- Two people want to go to a concert
- P1 prefers Bach, P2 Stravinsky

Bach or Stravinsky

- Two people want to go to a concert
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	B	S
B	2, 1	0, 0
S	0, 0	1, 2

Bach or Stravinsky

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- P1 prefers Bach, P2 Stravinsky

Correlated Equilibrium

- A *correlated joint policy* is a single distribution over the joint actions of all agents.
- A *correlated equilibrium* is a correlated joint policy where no agent i can increase their expected utility by deviating from the current policy.
- Easier to find than Nash equilibrium (Linear Program)

	B	S
B	2, 1	0, 0
S	0, 0	1, 2

~~$\pi^1(a^1)$~~ ~~$\pi^2(a^2)$~~ $\pi(a^1, a^2)$

$$\pi(B, B) = 0.5$$

$$\pi(S, S) = 0.5$$

[https://youtube.com/shorts/w3q77ZZIqwA
?si=J8H6L6W5kTRs-mUx](https://youtube.com/shorts/w3q77ZZIqwA?si=J8H6L6W5kTRs-mUx)

Recap

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Recap

- Games provide a mathematical framework for analyzing interaction between rational agents

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Recap

- Games provide a mathematical framework for analyzing interaction between rational agents
- Games may not have a single "optimal" solution; instead there are equilibria that may not be rankable

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- Every finite game has at least one Nash Equilibrium (pure or mixed)
- Mixed Nash equilibria occur when players are indifferent between two outcomes